Deep research request: why learners keep showing up — evidence catalogue for students in Świdnica, Poland and beyond

1. Developmental caveats (ages 10–14)

The cognitive, emotional, and psychosocial profile of a learner between the ages of 10 and 14—corresponding to the Polish lower-secondary school system, *klasy 4–8*—differs fundamentally from the adult populations upon which the majority of behavioural science, habit-formation, and goal-setting literature is based. Extrapolating findings from self-directed adults into this demographic without developmental translation introduces severe methodological and theoretical errors. This translation is especially critical in the contemporary Polish context, where the Ministry of National Education's April 2024 regulations effectively eliminated mandatory, graded homework in primary schools[1]. According to the Institute for Educational Research (IBE), this structural shift places the burden of academic practice entirely upon the learner's intrinsic motivation or the parents' supervisory capacity[1]. Understanding how a 12-year-old engages with voluntary mathematics practice requires recognizing their unique neurodevelopmental state.

During early adolescence, the brain undergoes a profound asynchronous maturation process. The striatal reward system, which drives novelty-seeking, social sensitivity, and responsiveness to immediate reinforcement, develops rapidly and peaks in reactivity during this stage. Conversely, the prefrontal cortex, which is responsible for executive functioning, delay of gratification, impulse control, and long-term planning, develops linearly and does not reach full maturity until the mid-twenties. Consequently, learners aged 10 to 14 are highly responsive to immediate cues and social rewards but possess highly fragile self-regulation capabilities. When operating alone, a 10-year-old cannot reliably act as their own commitment device.

Several foundational concepts in adult behavioural psychology require heavy modification when applied to this population. The concept of future-self continuity, for instance, allows adults to conceptualize a distal goal and use it to regulate present behaviour. Early adolescents possess a highly truncated temporal horizon; the abstract concept of the *egzamin ósmoklasisty* (eighth-grade exam) is too temporally distant for a 10-year-old to serve as a reliable intrinsic motivator. Furthermore, adult models of habit formation often presume an autonomous environment where an individual can engineer strict cue-routine-reward loops. A child's environment is not strictly autonomous; it is heavily mediated by parental schedules, sibling interference, and school timetables. True voluntary habits in children are rarely entirely self-initiated; they are heavily scaffolded by adults who provide the structural cue and enforce the routine until the behaviour stabilizes.

Additionally, metacognitive accuracy—the ability to assess one's own competence and calibrate effort accordingly—is notoriously underdeveloped at this age. Early adolescents frequently overestimate their mastery of a subject and underestimate the effort required to correct deficits. This is particularly dangerous in mathematics, a domain where anxiety often crystallizes during these exact years[2]. Meta-analytic evidence establishes a moderate, negative correlation ($r = -0.28$) between mathematics anxiety and academic achievement, a relationship that solidifies in late childhood and persists through adulthood[2]. Because working memory is heavily burdened by the affective distress of math anxiety, interventions that succeed in low-stress domains often backfire in mathematics.

Therefore, behavioural mechanics must separate the "learner alone" from the "learner with a parent." The solitary learner requires a system that functions as a prosthetic prefrontal cortex—supplying immediate, proximal goals, titrating difficulty dynamically to prevent anxiety-induced abandonment, and delivering near-instantaneous feedback.

2. The parent's role

When dealing with voluntary academic practice outside school hours, the presence of an adult fundamentally alters the psychological landscape. The empirical evidence distinguishes sharply between the mere quantity of parental involvement and the specific quality of that involvement. Overall, parental involvement in home learning yields a positive, albeit modest, association with academic adjustment across diverse demographic backgrounds[4]. However, the valence of this effect upon the child's motivation depends entirely on whether the parental approach is autonomy-supportive or controlling[6].

Autonomy-supportive parental involvement involves providing structure and scaffolding while acknowledging the child's perspective, offering meaningful choices, and providing rationales for why the practice is valuable without resorting to coercion. Evidence demonstrates that perceived parental autonomy support is positively associated with the student's autonomous homework motivation, productive behaviour, mathematics achievement, and self-regulated learning skills[5]. When parents act as an attentive audience and provide a stable structural loop—such as ensuring a quiet space and maintaining a consistent schedule—they effectively compensate for the child's developing executive functions. This supportive presence allows the child to internalize the value of the activity, moving along the continuum of self-determination toward intrinsic motivation.

Conversely, controlling parental involvement reliably undermines the child's motivation and learning outcomes[6]. Controlling behaviours are characterised by strict surveillance, unsolicited direct intervention in problem-solving, pressure to achieve specific grades, and the use of rigid reward-and-punishment contracts. Studies show that highly controlling parental behaviour exerts a negative impact on persistence, mathematics performance, and the development of independent study skills[5]. When a parent supplies high pressure or implements a transactional reward contract (e.g., paying for completed math modules), the child's locus of causality shifts entirely to the external environment. They perform the mathematics solely to appease the parent or secure the reward. This dynamic neutralizes any nascent intrinsic interest, heightens domain-specific anxiety, and ensures that the behaviour will immediately cease the moment the

parental pressure or reward is removed.
The evidence strictly cautions against parents utilizing digital dashboards primarily for punitive monitoring. When parental involvement focuses on person-based evaluations or normative comparisons, it actively damages the child's psychological need for relatedness and autonomy. Effective home environments separate the parent from the role of the enforcer, positioning them instead as a facilitator who removes environmental friction and supports the child's autonomous engagement.

3. Evidence catalogue

Field	**Details**
Principle	Sustained, high-quality voluntary engagement occurs only when an environment satisfies a learner's innate needs for autonomy (volition), competence (efficacy), and relatedness (social connection).
Evidence strength	Strong. Decades of research and massive recent meta-analyses confirm that satisfying these three psychological needs predicts positive affect, intrinsic motivation, and academic achievement, while frustrating them predicts disengagement.
Key studies	Vansteenkiste et al. (2020)[9]; Phillips (2021) Meta-analysis[11].
Effect size	Moderate to large. Autonomy satisfaction correlates with positive affect ($r = 0.43$) and negative affect ($r = -0.30$); competence satisfaction correlates with positive affect ($r = 0.45$) and negative affect ($r = -0.33$)[12].
Population tested	Broadly tested across all demographics globally. Explicitly tested on adolescents aged 10–14 in academic and mathematics-specific contexts[10].

Boundary conditions and failure modes	Providing choices that are overwhelming or irrelevant (false autonomy) induces anxiety rather than volition. If a mathematics task vastly exceeds a child's skill level, the need for competence is aggressively thwarted, leading to rapid abandonment of the activity. For the socially sensitive 10–14 age group, if academic practice isolates them from peers or results in social shaming (a threat to relatedness), they will actively avoid the platform.
Who has to do the work	The adult in the loop must calibrate the difficulty to ensure competence and frame the activity to support autonomy. If the learner is alone, the environment must automatically titrate difficulty and provide a genuine sense of volition.

Field	**Details**
Principle	Introducing highly salient, expected, and tangible extrinsic rewards for an activity that a learner already finds somewhat interesting will reliably decrease their intrinsic motivation to perform that activity once the reward is removed.
Evidence strength	Strong. Replicated across multiple major meta-analyses, though the effect is heavily dependent on the specific contingencies of the reward (the "undermining effect" or "overjustification effect").
Key studies	Deci, Koestner, & Ryan (1999)[13]; Tang & Hall (1995)[15]; Cerasoli et al. (2014)[17].
Effect size	Tang & Hall (1995) found engagement-contingent rewards

	undermined free-choice intrinsic motivation (d = -0.40) and completion-contingent rewards undermined it (d = -0.36)[15]. Cerasoli et al. found expected tangible rewards undermined free-choice intrinsic motivation (d = -0.41)[18].
Population tested	Extensively tested on children and college students in academic and laboratory settings. Tangible rewards tend to be significantly more detrimental to the intrinsic motivation of children than to adults[15].
Boundary conditions and failure modes	This effect applies primarily to tasks that already possess a baseline of intrinsic interest; one cannot undermine intrinsic motivation that does not exist (e.g., rote memorization for a deeply demotivated student). Unexpected rewards do not undermine motivation, nor do verbal rewards (praise) that are informational rather than controlling[15]. If a parent introduces a rigid contract, the child attributes their engagement entirely to the incentive, and practice ceases immediately when the incentive ends.
Who has to do the work	The adult must exercise restraint, resisting the urge to bribe the child for engagement, focusing instead on informational feedback. The unaccompanied learner is highly susceptible to reward mechanics.

Field	**Details**
Principle	Learners motivated by a desire to develop competence and deeply understand the material (mastery orientation) demonstrate greater persistence and resilience than learners motivated by a desire to outperform

	others or avoid looking incompetent (performance orientation).
Evidence strength	Strong. Consistent predictive validity in educational psychology, particularly regarding how students respond to academic failure and setbacks.
Key studies	Senko (2016); Urdan & Schoenfelder (2006); Lazarides & Watt (2015)[21].
Effect size	Moderate positive correlations between mastery goals and deep processing/persistence; moderate positive correlations between performance-avoidance goals and anxiety/surface learning.
Population tested	Highly validated in early adolescents (10–14 years old) specifically within mathematics classrooms.
Boundary conditions and failure modes	Performance-approach goals (wanting to be the best) can drive high short-term achievement in highly confident students, but they backfire catastrophically when a student encounters a setback, causing them to disengage entirely to protect their ego. For students with math anxiety, performance-avoidance goals (not wanting to look stupid) are paralyzing. Emphasizing normative grading or peer comparisons actively shifts students away from mastery and toward performance orientations.
Who has to do the work	The learner alone must internalize the value of the process. The adult must praise effort, strategy, and improvement rather than innate talent or relative ranking against peers.

Field	Details
Principle	A learner's likelihood of initiating and sustaining voluntary practice is a multiplicative function of their belief that they can succeed at the specific task (self-efficacy) and the subjective value they assign to the task.
Evidence strength	Strong. Self-efficacy is one of the most robust predictors of academic performance and persistence in the psychological literature.
Key studies	Bandura (1997); Huang (2013) Meta-analysis[24]; Kamsurya et al. (2022) Meta-analysis[26].
Effect size	Kamsurya et al. (2022) found a medium overall effect size ($g = 0.518$) of self-efficacy on mathematical abilities[26]. Math anxiety correlates moderately and negatively with math achievement ($r = -0.28$ to -0.32)[2].
Population tested	Extensively tested on children aged 10–14 in mathematics. Significant gender differences in math self-efficacy begin to emerge in late adolescence, favouring males despite equal aptitude[24].
Boundary conditions and failure modes	Self-efficacy is highly domain-specific. A child may have high self-efficacy in language arts but near-zero self-efficacy in geometry. If self-efficacy is artificially inflated through empty praise without underlying competence, the subsequent failure causes severe affective distress. Furthermore, if a task is perceived as having zero utility or intrinsic value, high self-efficacy will not trigger engagement. Math anxiety directly

	suppresses self-efficacy[29].
Who has to do the work	The environment or adult must structure "mastery experiences" (success on progressively harder tasks) to build the learner's genuine self-efficacy.

Field	Details
Principle	Setting specific, difficult, but attainable proximal (short-term) goals leads to higher engagement and performance than setting abstract, "do your best" goals or distal (long-term) goals.
Evidence strength	Strong for the general theory; Strong for the absolute necessity of proximal goals in children.
Key studies	Locke & Latham (2002)[30]; Bandura & Schunk (1981)[31].
Effect size	Meta-analyses show a small to medium positive unique effect of goal setting across behaviours (d = 0.34)[31]. Bandura & Schunk found massive persistence differences in children using proximal vs distal goals[33].
Population tested	Originally adult organizational behaviour, but explicitly tested and validated on primary and lower-secondary children in mathematics[33].
Boundary conditions and failure modes	Distal goals (e.g., "Study for the exam in two months") completely fail to motivate 10–14-year-olds because they lack the necessary future-self continuity and delay of gratification capabilities[33]. Furthermore, if a specific, difficult goal is assigned and the learner fails, it severely damages

	self-efficacy. Goal-setting works poorly for highly complex, novel, heuristic tasks where the learner does not yet know the strategies required; it works best for clearly defined practice.
Who has to do the work	The adult or the system must break down distal requirements into highly specific, immediate proximal goals (e.g., "Complete 5 algebra equations in the next 15 minutes").

Field	**Details**
Principle	Coupling a vivid visualization of a desired outcome and its internal obstacles (mental contrasting) with a specific if-then plan (implementation intention) significantly increases the likelihood of executing a voluntary behaviour.
Evidence strength	Moderate to Strong. Highly replicated in tightly controlled interventions, though effect sizes in unsupervised field settings are more modest.
Key studies	Duckworth et al. (2013)[36]; Oettingen (2012)[38].
Effect size	Generally yields small to medium effect sizes for academic goal attainment compared to goal-setting alone[39].
Population tested	Specifically tested and validated on middle-school children (ages 10–14) for academic outcomes and mathematics persistence[36].
Boundary conditions and failure modes	Mental contrasting fails if the goal is perceived as entirely unrealistic; the learner will discard the goal rather than form a plan.

	Implementation intentions ("If it is 16:00, then I will open my math book") require a stable environment to serve as a reliable cue. If the home environment is chaotic, the "if" trigger is continually disrupted. It requires accurate metacognition to identify the true obstacle.
Who has to do the work	The learner must actively engage in the cognitive exercise of identifying obstacles and forming the if-then plan, though an adult can guide the initial framework (WOOP).

Field	**Details**
Principle	Behaviour becomes automatic and requires less conscious willpower when it is repeatedly executed in a stable context triggered by a consistent cue, leading to a reliable reward.
Evidence strength	Moderate for academic study. Strong for simple motor tasks, but highly contested regarding the timeline and applicability to cognitively demanding tasks like mathematics.
Key studies	Lally et al. (2010); Wood & Neal (2007).
Effect size	Time to automaticity in adults ranges wildly from 18 to 254 days (median 66 days) for simple tasks; not reliably reported for childhood academic study.
Population tested	Predominantly self-directed adults focusing on health behaviours (diet, exercise). Downward transfer to 10–14-year-olds for academic practice is theoretically sound but empirically sparse without adult intervention.

Boundary conditions and failure modes	Studying mathematics is intrinsically cognitively demanding and emotionally fraught (if math anxiety is present). It is exceptionally difficult to render "doing math" into a purely automatic, non-conscious habit in the same way one automatically brushes one's teeth. Variations in homework load and shifting school timetables frequently destroy the stable context required for the cue to take hold.
Who has to do the work	The adult must engineer the environment to ensure the cue remains stable (e.g., same desk, same time) and prevent friction. The 10-year-old cannot reliably engineer their own contextual cues.

Field	**Details**
Principle	Tracking consecutive days of task completion leverages loss aversion, compelling the learner to engage in the behaviour even when unmotivated, simply to avoid the psychological pain of losing the accumulated streak.
Evidence strength	Moderate. The behavioural persistence effect is highly observable, but the psychological fallout is increasingly documented in the literature.
Key studies	Kahneman & Tversky (1979) on loss aversion; modern empirical analyses of educational technology platforms.
Effect size	High adherence rates while the streak is active; exact effect sizes are heavily platform-dependent and rarely reported in peer-reviewed independent academic

	literature.
Population tested	Broad digital populations of all ages. Downward transfer to 10–14-year-olds is highly relevant due to the prevalence of digital gaming mechanics in this cohort.
Boundary conditions and failure modes	The primary failure mode is catastrophic drop-off following a broken streak (the "what-the-hell" effect). If a 12-year-old builds a 40-day streak and misses a day due to illness, the accumulated loss aversion converts into acute demotivation. The streak becomes a measure of consecutive perfection rather than learning. Furthermore, learners will frequently engage in the lowest-effort, zero-learning action possible just to "save the streak," prioritizing the mechanic over the academic material.
Who has to do the work	The learner alone feels the internal psychological pressure of the streak, though the software system administers the tracking.

Field	**Details**
Principle	Adding game-design elements to non-game contexts (points, badges, leaderboards) increases short-term engagement and time-on-task through external reinforcement.
Evidence strength	Moderate. Gamification reliably boosts immediate engagement, but long-term effects on intrinsic motivation and deep learning are heavily contested and subject

	to severe novelty decay.
Key studies	Sailer & Homner (2020) Meta-analysis[40]; Dichev & Dicheva (2017).
Effect size	Sailer & Homner (2020) found medium to large effects on cognitive ($g = 0.82$), behavioural ($g = 0.74$), and affective outcomes ($g = 0.57$)[41].
Population tested	Wide range of students, including robust K-12 populations in STEM subjects[41].
Boundary conditions and failure modes	Points and badges (PBL) can crowd out intrinsic motivation if perceived as controlling[44]. Leaderboards are particularly dangerous for the 10–14 age group. While they motivate the top 10% of high performers, they actively demotivate and shame the bottom 80% who see mathematical evidence of their inferiority[44]. Gamification suffers from novelty decay; what motivates a student in week one becomes baseline expectation by week six, requiring ever-escalating rewards.
Who has to do the work	The environment/software administers the mechanics; the learner responds to the external stimuli.

Field	**Details**
Principle	Providing feedback directed at a learner's effort, strategy, and process fosters resilience, whereas providing feedback directed at their innate traits (e.g., "You are so smart at math") induces a fixed mindset and fragility in the face of failure.

Evidence strength	Contested / Moderate. The foundational laboratory logic holds, but large-scale field replications of mindset interventions show significantly smaller effects than originally claimed.
Key studies	Dweck (2006); Sisk et al. (2018) Meta-analysis[46]; Yeager et al. (2019)[46].
Effect size	Sisk et al. (2018) found a very small overall effect of growth mindset interventions on academic achievement (d = 0.05), which was non-significant when restricted to the highest-quality studies[47]. Yeager et al. (2019) found small positive effects primarily limited to previously lower-achieving students[46].
Population tested	Extensively tested on early adolescents and middle-schoolers (10–14) in mathematics.
Boundary conditions and failure modes	"Faux growth mindset" is a common failure mode, where an adult praises sheer effort even when the student is using an ineffective strategy and failing ("just try harder"). Process praise backfires if the child knows their performance is objectively poor, interpreting the praise as pity. Praise must be sincere, credible, and tied to specific strategic actions that lead to improvement.
Who has to do the work	The adult or the digital feedback system must meticulously frame praise around verbs (actions) rather than nouns (identities).

Field	**Details**

Principle	When a learner engages in a task where their skill level perfectly matches a high level of challenge, accompanied by clear goals and immediate feedback, they enter an intrinsically rewarding state of optimal experience and deep absorption.
Evidence strength	Contested in basic academic practice. While the theory is robust for experts, inducing it reliably in novice educational software is exceptionally difficult.
Key studies	Csikszentmihalyi (1990); Oliveira et al. (2021)[48].
Effect size	Oliveira et al. found that standard educational gamification had no statistically significant effect on students' flow experience ($r = 0.12$)[48].
Population tested	Broadly tested, but primarily validated in adults in creative, athletic, or high-expertise domains rather than children acquiring foundational skills.
Boundary conditions and failure modes	Flow requires the learner to have a requisite baseline of competence. A 12-year-old struggling with basic algebra cannot enter flow because the cognitive load of decoding the math prevents absorption. If the challenge exceeds skill by even a small margin in mathematics, the result is anxiety, not flow. If the challenge is slightly too low, the result is boredom. The "flow channel" for an average 10–14-year-old in voluntary mathematics is incredibly narrow.
Who has to do the work	The software must dynamically and flawlessly titrate difficulty on a granular level; otherwise, flow is impossible for the solitary learner.

Field	Details
Principle	Motivation, effort, and execution speed accelerate as a learner perceives themselves to be closer to completing a goal or reaching a reward.
Evidence strength	Strong. Rooted in foundational behaviourism and consistently replicated in consumer psychology and user experience design.
Key studies	Hull (1932)[49]; Kivetz, Urminsky, & Zheng (2006)[50]; Nunes & Drèze (2006)[49].
Effect size	Kivetz et al. (2006) observed a roughly 20% acceleration in action frequency as subjects approached a reward threshold[50].
Population tested	Primarily human adults in loyalty programs and behavioral economics studies[50]. Downward transfer to youth is highly plausible given the basic operant mechanics involved.
Boundary conditions and failure modes	The effect creates a predictable "post-reward slump." Immediately after crossing the finish line and obtaining the reward, motivation collapses and inter-action time increases dramatically[50]. Furthermore, if the goal is perceived as too distant, the gradient does not activate. The "endowed progress effect" (giving the user false initial progress) works to trigger the gradient earlier, but if users realize the progress is illusory, it triggers reactance[50].
Who has to do the work	The system must continuously frame long tasks as proximity to a finish line, and carefully structure the transitions between

	goals to mitigate the post-reward slump.

4. Claims that circulate widely — adjudicate each

The translation of behavioural science into commercial and educational sectors frequently results in the distortion of nuanced findings into absolute laws.

Growth Mindset as commonly sold: *Contested / Weak.* The popular narrative that simply teaching children "they can get smarter" will radically transform academic performance is not supported by rigorous data. The largest, pre-registered meta-analyses, such as Sisk et al. (2018), reveal an overall effect size nearing zero (d = 0.05)[47]. While it has marginal benefits for specific, low-achieving populations in highly supportive environments, it is not a panacea for motivation. Fostering mathematical competence yields a growth mindset, not merely the other way around.

21 Days to form a habit: *Folk.* There is no biological or psychological evidence that habits form in 21 days. The foundational Lally et al. (2010) study demonstrated a range of 18 to 254 days, heavily dependent on the complexity of the task. Voluntary mathematics practice is highly complex and emotionally laden; it will not become an automatic, thoughtless habit in three weeks.

The "Atomic Habits" framing: *Contested for children.* The concept of making habits tiny, obvious, attractive, and easy is fundamentally sound for adults who control their own environments. However, a 10–14-year-old does not control their schedule, their physical space, or their digital environment to the same degree. Expecting a child to self-engineer context cues without adult scaffolding is developmentally inappropriate.

Flow state as an achievable designed condition: *Weak in novice education.* While flow is a very real psychological phenomenon in experts, the claim that educational software can reliably place novice students into flow is largely marketing. The cognitive load of learning new mathematical concepts generally precludes the effortless absorption characteristic of flow, and studies confirm that standard gamification does not reliably induce it[48].

Gamification as a general motivator: *Moderate but transient.* Gamification is excellent for initiating behaviour and overcoming initial friction[40]. However, it relies entirely on extrinsic motivation. Over time, learners habituate to points and badges. Without an eventual transition to intrinsic value (competence, mastery), gamified engagement mechanics become a treadmill requiring ever-increasing stimuli, ultimately succumbing to novelty decay.

Dopamine-based explanations of motivation: *Folk neuroscience.* Popular literature frequently conflates dopamine with "pleasure" (e.g., "dopamine hits from badges"). Neurobiologically, dopamine is a neuromodulator related to *reward prediction error* and the salience of cues (wanting/craving), not the subjective experience of pleasure (liking). Designing for "dopamine hits" typically results in creating compulsive checking behaviours rather than durable, focused academic motivation.

5. Conflicts and open questions

Translating these varied psychological principles into a coherent software design or pedagogical strategy creates unavoidable theoretical and practical tensions. The principles do not exist in harmony; they frequently pull against one another.

Extrinsic Initiation vs. Intrinsic Undermining: A learner with zero initial motivation will not begin voluntary mathematics practice without a highly salient extrinsic reward, be it points, gamification mechanics, or parental contracts. However, self-determination theory and the overjustification effect warn that these exact extrinsic motivators will crowd out any chance of the student developing a genuine, intrinsic interest in the material[15]. How and when a system should "fade" extrinsic rewards to allow intrinsic motivation to take root without causing the student to churn remains an unsolved design challenge.

Streaks (Consistency) vs. The "What-the-hell" Effect (Resilience):
Streaks brilliantly leverage loss aversion to drive daily adherence. Yet, when a 12-year-old inevitably misses a day due to circumstances outside their control, the psychological punishment of the broken streak often causes total abandonment of the platform. The mechanic that most efficiently drives daily consistency directly undermines psychological resilience.

Competition vs. Demotivation: Leaderboards satisfy the need for competence and relatedness for the top 10% of high-achievers. However, achievement goal theory demonstrates that for the remaining 90%—particularly those grappling with math anxiety—normative comparisons force a performance-avoidance orientation. They log off to protect their self-esteem[44]. What motivates the strong actively discourages the weak.

Goal-Gradient Slump: Breaking learning into highly proximal goals triggers the goal-gradient effect, accelerating effort as the child nears the end of a module[50]. However, crossing that finish line triggers an immediate collapse in motivation[52]. Linking the end of one goal seamlessly to the beginning of another without causing user burnout is a persistent source of friction.

6. Ethics

When designing behavioural systems for children aged 10–14, strict ethical boundaries must be drawn between mechanics that increase *time-on-app* and mechanics that improve *learning*. Because the adolescent striatal system is highly responsive to variable rewards and social validation, it is remarkably easy to engineer compulsive engagement that yields zero educational value.

Illusory Progress and Deception: The endowed progress effect (giving a user artificial progress toward a goal to trigger the goal-gradient effect) walks a fine ethical line[50]. While it demonstrably increases adherence, if children realize the progress is manipulated or unearned, it breaches trust and violates the autonomy requirement of self-determination theory.

Variable Ratio Schedules:
Using intermittent, unpredictable rewards (the slot-machine mechanic) is highly effective at extending usage. However, this engages the child's compulsive reward-seeking circuitry without contributing to their mathematical competence. It trains the child to optimize their actions for the software's reward algorithms rather than mathematical understanding.

Punitive Loss Aversion: Leveraging loss aversion (e.g., "Your virtual pet will die if you don't practice math today") exploits a child's psychological vulnerabilities. It forces engagement through negative affect and anxiety. While they will show up to perform the task, they will do so under cognitive distress. This is particularly damaging in mathematics, where anxiety directly impairs the working memory capacity necessary for complex reasoning[2].

The ethical mandate for practitioner translation is to ensure that behavioural mechanics serve strictly as temporary scaffolding. The ultimate goal must be to build genuine mathematical competence and self-efficacy, rather than relying on permanent psychological crutches designed to monopolize a child's attention.

Works cited

1. Raport dotyczący prac domowych trafił do MEN - IBE PIB, https://ibe.edu.pl/pl/aktualnosci/3354-raport-dotyczacy-prac-domowych-trafil-do-men
2. A meta-analysis of the relation between math anxiety and ... - PubMed, https://pubmed.ncbi.nlm.nih.gov/33119346/
3. A Meta-Analysis of the Relation Between Math Anxiety and Math, https://www.researchgate.net/publication/346551018_A_meta-analysis_of_the_relation_between_math_anxiety_and_math_achievement
4. Full article: The effect of parental involvement on academic passion, https://www.tandfonline.com/doi/full/10.1080/02673843.2025.2467109
5. Parental homework involvement and students´achievement, https://portalinvestigacion.uniovi.es/documentos/662a9c55442b67790e19a604?lang=de
6. The Role of Parenting Styles in Schoolchildren's Homework, https://psychologyinrussia.com/volumes/?article=15065
7. Parental and Teacher Autonomy Support in Developing Self ... - MDPI, https://www.mdpi.com/2076-328X/15/12/1621
8. Reciprocal Relationships Between Parental Involvement ... - mijn-bsl, https://mijn.bsl.nl/reciprocal-relationships-between-parental-involvement-and-academ/50146692
9. Basic psychological need theory: Advancements, critical themes, https://www.researchgate.net/publication/338718454_Basic_psychological_need_theory_Advancements_critical_themes_and_future_directions
10. Basic psychological needs in the classroom - Universidad de Murcia, https://digitum.um.es/bitstreams/79f6ed11-2424-4c31-bc96-abda72d31fc2/download
11. (PDF) A meta-analytic investigation of the relationship between, https://www.researchgate.net/publication/350938414_A_meta-analytic_investigation_of_the_relationship_between_basic_psychological_need_satisfaction_and_affect
12. A Meta-Analytic Investigation of the Relationship between Basic, https://rune.une.edu.au/bitstreams/b3714725-52c2-49fa-80ac-f9dae8e67cc7/download
13. Overjustification Effect: Why Rewards Backfire - Yu-kai Chou, https://yukaichou.com/behavioral-analysis/overjustification-effect-lepper-greene-int

rinsic-motivation/
14. Continuing the Controversy About Reward and Intrinsic Motivation, https://www.ovid.com/journals/psycc/fulltext/10.1037/004787~continuing-the-controversy-about-reward-and-intrinsic
15. A Meta-Analytic Review of Experiments Examining the Effect of, https://albimarketing.com/docs/A%20Meta-Analytic%20Review%20of%20Experiments%20Examining%20the%20Effect%20of%20Extrinsic%20Rewards%20on%20Intrinsic%20Motivation.pdf
16. The overjustification effect: A meta‐analysis - Semantic Scholar, https://www.semanticscholar.org/paper/The-overjustification-effect%3A-A-meta%E2%80%90analysis-Tang-Hall/bcf84e2e08b82cb62836a3e2edc68a6ad53f35e2
17. (PDF) The undermining effect revisited: The salience of everyday, https://www.researchgate.net/publication/281488990_The_undermining_effect_revisited_The_salience_of_everyday_verbal_rewards_and_self-determined_motivation
18. Intrinsic vs. Extrinsic Motivation Impact | PDF - Scribd, https://www.scribd.com/document/967003436/2014-03897-001
19. When do extrinsic rewards undermine intrinsic motivation? A meta, https://www.researchgate.net/publication/367220662_When_do_extrinsic_rewards_undermine_intrinsic_motivation_A_meta-analysis
20. Overjustification effect - Wikipedia, https://en.wikipedia.org/wiki/Overjustification_effect
21. The role of classroom characteristics for students' motivation and, https://www.semanticscholar.org/paper/The-role-of-classroom-characteristics-for-students%E2%80%99-Lazarides-Rohowski/bd2ae02366d6b7de8e3e387b030e7481150efeaf
22. Teacher Enthusiasm, Supportive Instructional Practices, and Student, https://www.researchgate.net/publication/354725343_Teacher_Enthusiasm_Supportive_Instructional_Practices_and_Student_Motivation_in_Mathematics_Classrooms
23. Chapter 19 Investigating Mathematics Students' Motivations and, https://brill.com/view/book/9789004745308/BP000039.xml
24. Gender differences in academic self-efficacy: a meta-analysis, https://www.researchgate.net/publication/257488399_Gender_differences_in_academic_self-efficacy_a_meta-analysis
25. Gender, self-efficacy and performance in a mathematics test - SciELO, https://www.scielo.sa.cr/scielo.php?pid=S2215-34702022000100722&script=sci_arttext&tlng=en
26. The Effect of Self-Efficacy on Students' Mathematical Abilities, https://www.researchgate.net/publication/359979767_The_Effect_of_Self-Efficacy_on_Students'_Mathematical_Abilities_A_Meta-Analysis_Study
27. The Effect of Self-Efficacy on Students' Mathematical Abilities: A Meta, https://jpp.fkip.unila.ac.id/index.php/jpp/article/download/469/324
28. How Does Math Anxiety Affect Students' Academic Achievement in, https://www.icck.org/article/html/jsspa.2024.672218

29. The relationship between students' self-efficacy and mathematics, https://www.academia.edu/98962294/The_relationship_between_students_self_efficacy_and_mathematics_anxiety_Meta_analysis_investigation
30. refinement is the science and practice of goal setting theory in, https://www-2.rotman.utoronto.ca/facbios/file/04%20-%20Latham%20CAN%20Psych%202001.pdf
31. Unique effects of setting goals on behavior change - SciSpace, https://scispace.com/pdf/unique-effects-of-setting-goals-on-behavior-change-4vnder4wzg.pdf
32. (PDF) The effect of self-efficacy and self-set grade goals on, https://www.researchgate.net/publication/379777100_The_effect_of_self-efficacy_and_self-set_grade_goals_on_academic_outcomes
33. Effect of Goal Proximity and Specificity on Endurance Performance, https://journals.humankinetics.com/downloadpdf/journals/jsep/10/1/article-p81.pdf
34. Unique Effects of Setting Goals on Behavior Change - eClass, https://eclass.uth.gr/modules/document/index.php?course=PE_P_160&download=/5deb6647ojG3/5f796bb3xBMv.pdf
35. Exploring the Moderating Effect of Self-Efficacy - SJSU ScholarWorks, https://scholarworks.sjsu.edu/cgi/viewcontent.cgi?article=8861&context=etd_theses
36. From Fantasy to Action: Mental Contrasting With Implementation, https://www.researchgate.net/publication/256294319_From_Fantasy_to_Action_Mental_Contrasting_With_Implementation_Intentions_MCII_Improves_Academic_Performance_in_Children
37. Mental Contrasting with Implementation Intentions to Lower Test, https://scholarsarchive.byu.edu/cgi/viewcontent.cgi?article=10165&context=etd
38. WOOP Mental Contrasting: Oettingen's 4-Step Method - Yu-kai Chou, https://yukaichou.com/gamification-analysis/woop-mental-contrasting-oettingen-wish-outcome-obstacle-plan/
39. (PDF) A Meta-Analysis of the Effects of Mental Contrasting With, https://www.researchgate.net/publication/351524219_A_Meta-Analysis_of_the_Effects_of_Mental_Contrasting_With_Implementation_Intentions_on_Goal_Attainment
40. Game-Based and Gamified Robotics Education - arXiv, https://arxiv.org/html/2601.22199v1
41. A meta-analysis on the influence of gamification in formal, https://www.researchgate.net/publication/353815817_A_meta-analysis_on_the_influence_of_gamification_in_formal_educational_settings_on_affective_and_behavioral_outcomes
42. Benefits of Gamified Learning in Physical Education Students, https://revista-apunts.com/en/benefits-of-gamified-learning-in-physical-education-students-a-systematic-review/
43. Examining the effectiveness of gamification as a tool promoting, https://pmc.ncbi.nlm.nih.gov/articles/PMC10591086/
44. Gamification in Qur'anic Learning: Evidence from Muslim Students in,

https://journals.ldpb.org/index.php/cognoscere/article/download/285/247
45. A Systematic Literature Review of Gamification in Education, https://pdfs.semanticscholar.org/6fad/fd747349d307d4eb9ca79e5adcb7f3a0ee47.pdf
46. Growth Mindset and Academic Resilience as Predictors of, https://www.eduresearchjournal.com/index.php/ijei/article/download/528/524/1302
47. (PDF) Do Growth Mindset Interventions Impact Students' Academic, https://www.researchgate.net/publication/365261074_Do_Growth_Mindset_Interventions_Impact_Students'_Academic_Achievement_A_Systematic_Review_and_Meta-Analysis_with_Recommendations_for_Best_Practices
48. The Effects of Gamification on Students' Flow Experience, https://cris.tuni.fi/ws/portalfiles/portal/158562826/The_Effects_of_Gamification_on_Students_Flow_Experience.pdf
49. Goal Gradient Effect: Speed up user progress - Learning Loop, https://learningloop.io/plays/psychology/goal-gradient-effect
50. Goal-Gradient Hypothesis: Why We Speed Up at the End, https://yukaichou.com/behavioral-analysis/goal-gradient-hypothesis-hull-kivetz-motivation-acceleration/
51. The Goal-Gradient Hypothesis Resurrected: Purchase Acceleration, https://www.researchgate.net/publication/239776073_The_Goal-Gradient_Hypothesis_Resurrected_Purchase_Acceleration_Illusionary_Goal_Progress_and_Customer_Retention
52. The Impact of Successful Reward Attainment on Purchase Behavior, https://www.researchgate.net/publication/259888864_Recurring_Goals_and_Learning_The_Impact_of_Successful_Reward_Attainment_on_Purchase_Behavior
53. (PDF) It's About Time: Earlier Rewards Increase Intrinsic Motivation, https://www.researchgate.net/publication/325363397_It's_about_time_Earlier_rewards_increase_intrinsic_motivation
54. Math Anxiety's Impact on Performance | PDF | Effect Size | Psychology, https://www.scribd.com/document/745856113/fpsyg-12-798090